
Deep Variational Frailty Models for Survival Prediction Under Dependent Censoring

Abstract

Most survival models assume independent censoring: after conditioning on observed covariates, the censoring process carries no additional information about the event time. This assumption can be violated in clinical, longitudinal, and reliability settings, where unobserved factors may jointly affect failure risk and dropout or removal from follow-up. Because dependent censoring is not identifiable from right-censored data without structural assumptions, improving prediction in this setting requires an explicit inductive bias. We propose the Deep Variational Frailty Model (DVFM), a latent-variable framework that models event and censoring times as conditionally independent given observed covariates and a shared latent frailty variable. Marginalizing over this frailty induces flexible dependence between the event and censoring processes while avoiding the need to choose a fixed copula family in advance. We derive a censored evidence lower bound for right-censored observations and optimize it using amortized variational inference. DVFM should therefore be interpreted not as a test for dependent censoring, but as a structured predictive model for settings where shared latent heterogeneity is plausible. Experiments on synthetic benchmarks with varying dependence mechanisms, together with evaluations on real survival datasets, show that DVFM often improves discrimination and calibration relative to standard deep survival baselines and a representative parametric copula baseline, especially on survival-oriented metrics under dependent censoring.

1 Introduction

Survival analysis studies time-to-event outcomes in domains such as healthcare (time to relapse or death), engineering (time to failure), and finance (time to default) [1]. A defining characteristic is *censoring*: for many individuals, the event time is not observed within the study window. Standard estimators and a large fraction of modern deep survival models rely on the *independent censoring* assumption:

$$E \perp C \mid X, \quad (1)$$

where E denotes event time, C censoring time, and X observed covariates. In many real settings, Eq. (1) is violated. For instance, patients experiencing severe side effects or deterioration are more likely to withdraw early; similarly, high-risk devices may be removed from monitoring. Such *dependent (informative) censoring* introduces systematic bias: models trained assuming independence can overestimate or underestimate survival for censored individuals and miscalibrate predicted risk.

Dependent censoring is challenging not only empirically but also theoretically due to *non-identifiability*. With right-censoring, we observe only $T = \min(E, C)$ and $\delta = \mathbb{I}(E \leq C)$. Tsiatis showed that without structural assumptions, the joint distribution of (E, C) is not identifiable from the distribution of (T, δ) alone [2]. Therefore, addressing dependent censoring necessarily requires an *inductive bias*.

Classical approaches introduce inductive bias using copulas: they assume a parametric dependence family (e.g., Clayton, Gumbel, Frank) combining parametric marginals over event times and censoring

times [3]. Copulas provide a coherent likelihood under the assumed dependence but require choosing a dependence family a priori. This is problematic in complex, high-dimensional settings where dependence geometry may be unknown, heterogeneous, or covariate-dependent.

We propose a different inductive bias grounded in the *shared frailty* view: dependence between E and C arises from unobserved heterogeneity, such as latent health status or latent failure propensity. We operationalize this idea via a deep latent-variable model, learning the latent heterogeneity with amortized variational inference. This yields **Deep Variational Frailty Models (DVFM)**, a joint generative framework for event and censoring processes that learns an implicit dependence structure from data.

Contributions.

1. **Model.** We introduce DVFM, a deep generative framework that models $p(E, C \mid \mathbf{X})$ using a shared latent frailty Z , yielding a flexible, data-driven dependence mechanism without requiring an a priori specific copula family.
2. **Objective.** We derive a censored ELBO (C-ELBO) for right-censored observations and show how it enables end-to-end optimization of the joint likelihood using amortized inference.
3. **Theory.** We incorporate the Tsiatis [2] nonidentifiability challenge directly into the modeling story and provide a structural identifiability discussion: within a meaningful frailty-restricted subclass, the model is well posed; more broadly, the DVFM inductive bias acts as a structural filter that rules out proxy solutions that fit observed likelihood but violate the model class.

2 Related Work

Modern survival analysis. DeepSurv [4] extends Cox proportional hazards [5] with neural risk functions, while discrete-time approaches such as MTLR [6] and DeepHit [7] directly model survival probabilities over time. These methods typically assume independent censoring, either explicitly through partial likelihood construction or implicitly through losses that do not model the censoring mechanism.

Dependent censoring and copulas. Copula models represent joint distributions through marginal CDFs and a copula function [8]. In dependent censoring, a copula provides a parametric bridge between E and C with a global dependence parameter. While theoretically elegant, performance depends on selecting an appropriate family; misspecification (e.g., lower-tail vs upper-tail dependence) can lead to biased inference.

Frailty models. Frailty introduces random effects into hazard functions to account for unobserved heterogeneity; shared frailty links multiple times (e.g., event and censoring) through a common latent factor [9, 10]. Classical frailty models often assume Gamma or log-normal frailty and rely on EM-type estimation. DVFM generalizes this by learning non-linear mappings and using amortized variational inference [11].

Deep generative models for survival. Generative survival models (e.g., mixture-based or VAE-like approaches) commonly model the event-time marginal $p(E \mid \mathbf{X})$ but do not explicitly model the censoring mechanism jointly. DVFM’s central design is to model the *joint* generative process for both event and censoring and to learn dependence via shared latent frailty.

3 Preliminaries and Problem Definition

Let $E \in \mathbb{R}_+$ be event time, $C \in \mathbb{R}_+$ censoring time, and $\mathbf{x} \in \mathbb{R}^p$ covariates. Observed data are $(t, \delta, \mathbf{x})$ with

$$t = \min(E, C), \quad \delta = \mathbb{I}(E \leq C). \quad (2)$$

We seek to estimate the conditional event survival curve $S_E(e \mid \mathbf{x}) = \Pr(E > e \mid \mathbf{x})$ and accurate conditional survival prediction for censored subjects (i.e., predictions for E given that only $E > t$ is observed).

84 3.1 The independent censoring assumption

85 Under independent censoring, the observed-data likelihood factorizes into separate event and censor-
 86 ing terms. Writing f_E, S_E and f_C, S_C for the event and censoring densities and survival functions,
 87 respectively, the joint likelihood over a dataset $\mathcal{D} = \{(t_i, \delta_i, x_i)\}_{i=1}^N$ is

$$\mathcal{L}(\mathcal{D}) = \prod_{i=1}^N [f_E(t_i | x_i)^{\delta_i} S_E(t_i | x_i)^{1-\delta_i}] [f_C(t_i | x_i)^{1-\delta_i} S_C(t_i | x_i)^{\delta_i}]. \quad (3)$$

88 Because the censoring term does not involve event parameters, many methods discard it and optimize
 89 only the event component. When $E \not\perp C | X$, maximizing the simplified event-only likelihood is
 90 generally biased. The deeper issue is non-identifiability: only (T, δ) is observed, so multiple joint
 91 models for (E, C) can induce the same observed distribution.

92 3.2 The Tsiatis non-identifiability challenge

93 Determining whether the independent censoring assumption holds in a given dataset is extremely
 94 difficult. The fundamental problem is that we never simultaneously observe both the variables E
 95 and C for any subject, we only observe the follow-up time $T = \min(E, C)$ and an event indicator
 96 $\delta = \mathbb{I}(E \leq C)$. This leads to the Tsiatis nonidentifiability theorem: for any dependent joint
 97 distribution of E and C , there exists an independent proxy distribution P^* that matches the observed
 98 data perfectly [2]. Because the observed data are consistent with both a dependent reality and an
 99 independent proxy, the likelihood function cannot distinguish between them. Standard survival
 100 models tend to converge to this independent proxy. This behavior hides the true dependency and
 101 leads to bias in the prediction of the survival distribution.

102 To overcome this barrier, one must introduce *structural constraints* that favor the true marginal
 103 distributions over the distorted marginals required by the Tsiatis proxy. In our setting, that structure
 104 is the latent frailty model

$$E \perp C | (X, Z), \quad (4)$$

105 which imposes conditional independence at the latent level while allowing flexible marginal depen-
 106 dence after integrating out z .

107 4 Method: Deep Variational Frailty Model

108 DVFM assumes that dependence arises through a shared latent frailty variable $Z \in \mathbb{R}^d$ such that (4)
 109 holds. The joint generative model is

$$p_\theta(e, c, z | x) = p_\theta(e | x, z) p_\theta(c | x, z) p(z), \quad (5)$$

110 with prior $p(z) = \mathcal{N}(0, I)$. The latent variable z represents unobserved heterogeneity such as
 111 latent health status, frailty, or failure propensity. Although E and C are conditionally independent
 112 given (X, Z) , marginalizing z induces dependence between them. Figure 1 summarizes the DVFM
 113 architecture.

114 4.1 Neural parameterization of event and censoring

115 A decoder network outputs the distributional parameters of the event and censoring models:

$$[\psi_E(x, z), \psi_C(x, z)] = \text{Decoder}_\theta(x, z). \quad (6)$$

116 In our implementation, both event and censoring times follow Weibull distributions:

$$E | x, z \sim \text{Weibull}(k_E(x, z), \lambda_E(x, z)), \quad (7)$$

$$C | x, z \sim \text{Weibull}(k_C(x, z), \lambda_C(x, z)). \quad (8)$$

118 Positivity of parameters is enforced via softplus activations. The Weibull survival and density
 119 functions are

$$S(t | k, \lambda) = \exp \left[- \left(\frac{t}{\lambda} \right)^k \right], \quad f(t | k, \lambda) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} \exp \left[- \left(\frac{t}{\lambda} \right)^k \right]. \quad (9)$$

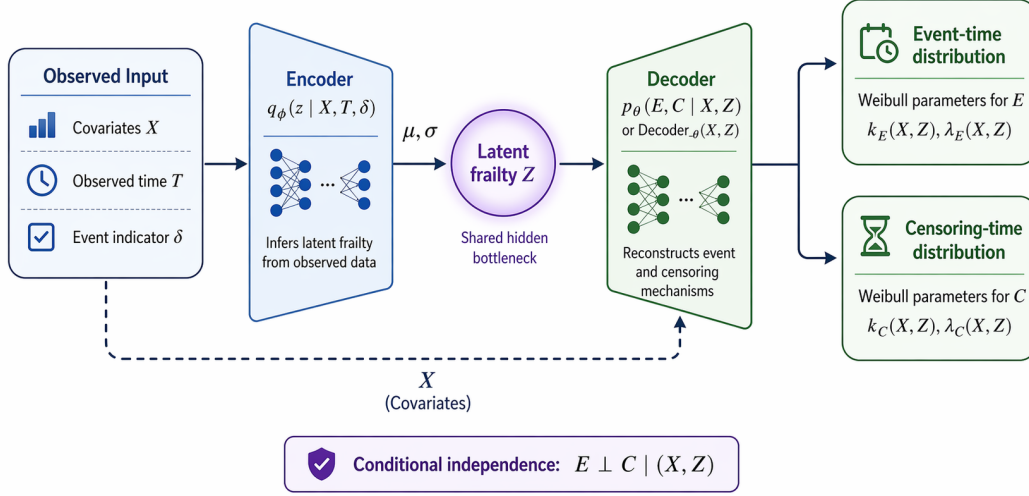


Figure 1: Architecture of the Deep Variational Frailty Model (DVFM). The encoder infers a latent frailty variable z from the observed tuple (x, t, δ) , and the decoder combines (x, z) to generate the parameters of the event-time and censoring-time distributions. Conditional on (x, z) , the event and censoring times are modeled as independent, while marginal dependence is induced through the shared latent frailty.

4.2 Amortized variational inference

The posterior $p(z | t, \delta, x)$ is intractable, so we introduce an encoder network defining a Gaussian variational posterior

$$q_\phi(z | t, \delta, x) = \mathcal{N}(z; \mu_\phi(t, \delta, x), \text{diag}(\sigma_\phi^2(t, \delta, x))). \quad (10)$$

We train the model using the reparameterization trick, writing $z = \mu_\phi + \sigma_\phi \odot \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$.

4.3 Censored evidence lower bound (C-ELBO)

Under (4), the observed likelihood contribution of one instance (t, δ, x) given (x, z) is

$$\mathcal{L}_{\text{joint}}(t, \delta | x, z) = [f_E(t | x, z) S_C(t | x, z)]^\delta [S_E(t | x, z) f_C(t | x, z)]^{1-\delta}. \quad (11)$$

The C-ELBO is then

$$\mathcal{L}_{\text{C-ELBO}}(\theta, \phi) = \mathbb{E}_{q_\phi(z | t, \delta, x)} [\log \mathcal{L}_{\text{joint}}(t, \delta | x, z)] - \beta D_{\text{KL}}(q_\phi(z | t, \delta, x) \| p(z)). \quad (12)$$

We anneal β from 0 to 1 (or to a capped β_{max}) and optionally apply *free bits* per latent dimension to mitigate posterior collapse [11].

Why joint modeling matters? Independence-based methods effectively discard the censoring model. DVFM treats censoring as informative and uses the censoring likelihood term to learn shared latent structure. This is particularly important for counterfactual survival prediction of censored subjects, where independence-based training is often over-optimistic.

4.4 Prediction via latent marginalization

At test time, for a new subject we typically observe only covariates x , while (t, δ) are unobserved. Since DVFM represents subject-specific heterogeneity through a latent frailty variable z , the event survival function is obtained by marginalizing out z :

$$S_E(e | x) = \Pr(E > e | x) = \int S_E(e | x, z) \pi(z) dz, \quad (13)$$

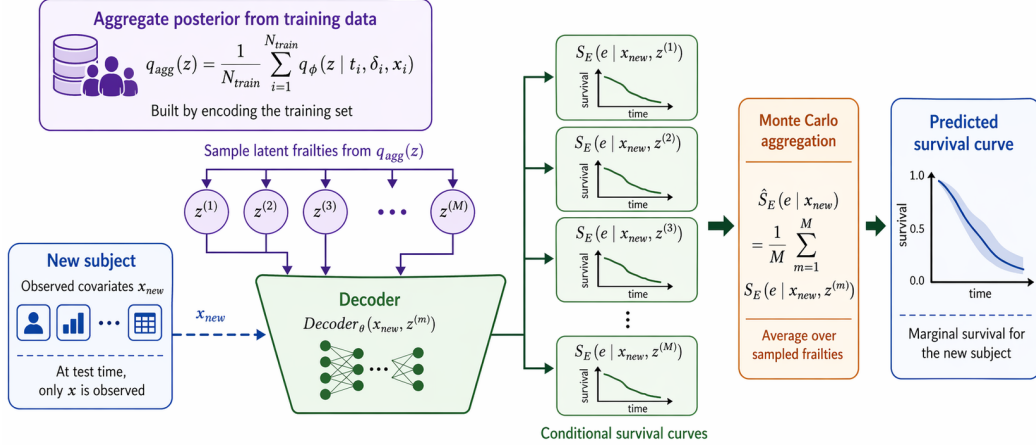


Figure 2: Prediction for a new subject in DVFM via latent marginalization. At test time, only the covariates x_{new} are observed. Latent frailty samples are drawn from the aggregate posterior $q_{agg}(z)$, passed through the decoder together with x_{new} , and used to generate conditional survival curves $S_E(e | x_{new}, z^{(m)})$. The final prediction is obtained by Monte Carlo averaging over these sampled frailties, yielding the marginal survival curve for the new subject.

where $S_E(e | x, z)$ is the conditional survival function induced by the decoder. In our Weibull parameterization,

$$S_E(e | x, z) = \exp\left(-\left(\frac{e}{\lambda(x, z)}\right)^{k(x, z)}\right), \quad (14)$$

with $[k(x, z), \lambda(x, z)] = \text{Decoder}_{\theta}(x, z)$.

Because the integral in Eq. (13) is not available in closed form, we approximate it by Monte Carlo averaging:

$$\hat{S}_E(e | x) = \frac{1}{M} \sum_{m=1}^M S_E(e | x, z^{(m)}), \quad z^{(m)} \sim \pi(z), \quad (15)$$

where M is the number of draws. We consider two choices for the sampling distribution $\pi(z)$. The first is the prior, $\pi(z) = p(z) = \mathcal{N}(0, I)$. The second is the aggregate posterior,

$$q_{agg}(z) = \frac{1}{N_{train}} \sum_{i=1}^{N_{train}} q_{\phi}(z | t_i, \delta_i, x_i), \quad (16)$$

which better reflects the latent regions supported by the training data. Under aggregate-posterior sampling, we first encode the training set to obtain posterior parameters $\{(\mu_i, \sigma_i)\}_{i=1}^{N_{train}}$, then for each Monte Carlo draw, we choose an index j uniformly from the training set and draw

$$z^{(m)} \sim \mathcal{N}(\mu_j, \text{diag}(\sigma_j^2)). \quad (17)$$

The final prediction is again obtained by Eq. (15). In practice, aggregate-posterior marginalization is often more robust than prior sampling when the learned posterior differs noticeably from the prior, because it restricts prediction-time sampling to latent regions actually visited during training. Figure 2 illustrates the prediction procedure for a new subject.

5 Modeling Dependent Censoring: Copulas vs. Latent Frailty

5.1 Copula-based dependent censoring

Copula models represent the joint distribution through marginals and a copula function:

$$F_{E,C}(E, C | X) = C_{\eta}(F_E(E | X), F_C(C | X)), \quad (18)$$

where F_E, F_C are marginal CDFs and C_{η} is a copula parameterized by η . This imposes an explicit global dependence structure. In practice, the correct family may be unknown, and the geometry of dependence may vary across covariate strata.

157 5.2 DVFM as an implicit copula via latent marginalization

158 Under (4), the joint survival function under DVFM is

$$S(e, c | x) = \int S_E(e | x, z) S_C(c | x, z) p(z) dz. \quad (19)$$

159 This need not factorize into $S_E(e | x) S_C(c | x)$, so the model induces dependence without fixing a
160 copula family in advance.

161 **Proposition 1** (Latent frailty induces dependent censoring). *Assume $E \perp C | (X, Z)$, and both*
162 *$S_E(\cdot | x, z)$ and $S_C(\cdot | x, z)$ depend non-trivially on the same latent variable z . Then, in general,*

$$S(e, c | x) \neq S_E(e | x) S_C(c | x), \quad (20)$$

163 *and hence E and C are marginally dependent given x .*

164 *Proof.* From (19), write $S(e, c | x) = \int f(z)g(z)p(z)dz$, where $f(z) = S_E(e | x, z)$ and $g(z) =$
165 $S_C(c | x, z)$. In general, $\int fg dp \neq (\int f dp)(\int g dp)$. Since both terms depend non-trivially on z ,
166 factorization fails, implying marginal dependence. $\square$

167 **Conceptual comparison.** Copulas impose dependence *top-down* via an explicit family. DVFM
168 learns dependence *bottom-up* through latent heterogeneity. This provides flexibility, covariate
169 adaptation, and reduced brittleness to misspecification.

170 5.3 What Can Be Identified Under a Restricted Frailty Structure

171 The Tsiatis result rules out unrestricted identification of the joint law of (E, C) from right-censored
172 observations alone. The purpose of the following result is therefore modest: it shows that, after
173 imposing a strong multiplicative frailty structure and sufficient variation in the hazard indices, the
174 induced joint survival function is identifiable within that restricted class. This result should be read as
175 a structural sanity check, not as a general identifiability theorem for the full neural DVFM.

176 **Assumption 1** (Multiplicative frailty structure). *For fixed covariates x , suppose*

$$S_E(e | x, z) = \exp\{-\Lambda_E(e, x)\phi(z)\}, \quad S_C(c | x, z) = \exp\{-\Lambda_C(c, x)\psi(z)\},$$

177 *where Λ_E and Λ_C are nonnegative cumulative hazard indices and $\phi(z), \psi(z) > 0$.*

178 **Assumption 2** (Fixed frailty law). *The prior law of Z is fixed, and the induced law of $(\phi(Z), \psi(Z))$*
179 *does not depend on x .*

180 **Assumption 3** (Sufficient hazard-index variation). *The set*

$$\{(\Lambda_E(t, x), \Lambda_C(t, x)) : t \geq 0, x \in \mathcal{X}\}$$

181 *contains a nonempty open subset of $\mathbb{R}_+^2$, and the hazard indices are known or identified within the*
182 *restricted model class.*

183 **Theorem 1** (Restricted identifiability of the induced joint survival). *Under Assumptions 1–3, the*
184 *observed all-cause survival function*

$$S_{\text{obs}}(t | x) = \Pr(T > t | x)$$

185 *identifies the bivariate Laplace transform of $(\phi(Z), \psi(Z))$ on an open subset of $\mathbb{R}_+^2$. Consequently,*
186 *the induced joint survival function*

$$S(e, c | x) = \mathbb{E}_Z [\exp\{-\Lambda_E(e, x)\phi(Z) - \Lambda_C(c, x)\psi(Z)\}]$$

187 *is identifiable within this restricted multiplicative frailty class.*

188 *Proof.* Under the multiplicative frailty model,

$$S_{\text{obs}}(t | x) = \Pr(E > t, C > t | x) = \mathbb{E}_Z [\exp\{-\Lambda_E(t, x)\phi(Z) - \Lambda_C(t, x)\psi(Z)\}].$$

189 Define the bivariate Laplace transform

$$L(a, b) = \mathbb{E}_Z [\exp\{-a\phi(Z) - b\psi(Z)\}].$$

Metric	DeepSurv	MTLR	ClaytonAFT	DVFM
C-index $\uparrow$	0.781 $\pm$ 0.054	0.759 $\pm$ 0.054	0.857 $\pm$ 0.039	0.863 $\pm$ 0.034
IBS (IPCW) $\downarrow$	0.089 $\pm$ 0.013	0.122 $\pm$ 0.024	0.081 $\pm$ 0.012	0.074 $\pm$ 0.012
IBS-DEP $\downarrow$	0.072 $\pm$ 0.033	0.082 $\pm$ 0.039	0.059 $\pm$ 0.026	0.051 $\pm$ 0.023
MAE Uncensored $\downarrow$	0.385 $\pm$ 0.138	0.390 $\pm$ 0.177	0.337 $\pm$ 0.120	0.334 $\pm$ 0.122
MAE Censored $\downarrow$	2.634 $\pm$ 1.401	2.780 $\pm$ 1.433	1.753 $\pm$ 0.918	2.026 $\pm$ 1.105

Table 1: Comparison across standard metrics, dependence-aware calibration, and bias decomposition on the synthetic benchmark (mean $\pm$ std over 5 runs).

Scenario	Copula	Dependence	θ	Censoring Rate	True τ
S1	Clayton	low	1	0.4379	0.3525
S2	Clayton	high	10	0.4329	0.8332
S3	Frailty-Mixture	low	0.5	0.5458	0.0153
S4	Frailty-Mixture	high	10	0.5511	0.3800
S5	Frank	low	0.2	0.3894	0.0289
S6	Frank	high	10	0.3926	0.3557
S7	Gaussian	low	0.1	0.4304	0.0197
S8	Gaussian	high	0.8	0.4313	0.3412
S9	Gumbel	low	1.1	0.4697	0.0172
S10	Gumbel	high	10	0.4699	0.3638

Table 2: Synthetic scenarios used in the benchmark.

190 Then

$$S_{\text{obs}}(t \mid x) = L(\Lambda_E(t, x), \Lambda_C(t, x)).$$

191 By Assumption 3, the observed survival function evaluates $L(a, b)$ on a nonempty open subset of
192 $\mathbb{R}_+^2$. Since Laplace transforms of nonnegative random variables are analytic and uniquely determine
193 the corresponding distribution, the law of $(\phi(Z), \psi(Z))$ is identified within this restricted class. The
194 displayed expression for $S(e, c \mid x)$ is therefore identified. $\square$

195 **Remark 1.** *This theorem does not imply that the full neural DVFM is identifiable. It only states that*
196 *a strongly restricted multiplicative frailty model can be identifiable when the hazard-index structure*
197 *is known or otherwise identified and varies sufficiently. In the full DVFM, the frailty representation*
198 *should be interpreted as an inductive bias for prediction rather than as a guarantee of recovering the*
199 *true joint law of (E, C) .*

200 6 Why DVFM Works Under Both Independent and Dependent Censoring

201 DVFM is designed for prediction: it aims to improve recovery of the event-time distribution when
202 censoring is dependent, while remaining competitive when the standard independent-censoring
203 assumption holds. This behavior can be understood through the C-ELBO objective (12), which
204 balances accurate joint reconstruction against latent complexity.

205 Under independent censoring, the observed data can already be explained without introducing shared
206 latent dependence. In this regime, the reconstruction term does not require the posterior to encode
207 additional dependence information through the frailty variable, and the KL penalty discourages
208 unnecessary deviations from the prior. As a result, DVFM can behave similarly to a standard survival
209 predictor and need not lose performance.

210 Under dependent censoring, however, an independence-based explanation may fit the observed data
211 only through the Tsatis proxy, which can distort the event and censoring mechanisms. DVFM instead
212 explains dependence through conditional independence given a shared latent frailty,

$$E \perp C \mid (X, Z),$$

213 allowing the model to improve the joint reconstruction term by capturing shared heterogeneity.
214 When this reconstruction gain outweighs the additional KL cost, optimization can favor a nontrivial
215 latent representation, which may improve prediction under informative censoring. This effect is

Dataset	#Instance	#Features	Censoring Rate
ALS PRO-ACT	8883	26	0.733
Cancer METABRIC	1981	79	0.552
ALS CALSNIC	94	22	0.340

Table 3: Real dataset statistics. Censoring rates are computed over the full dataset.

Metric	CoxPH	DeepSurv	MTLR	ClaytonAFT	DVFM
C-index $\uparrow$	0.746 ± 0.041	0.733 ± 0.032	0.801 ± 0.028	0.785 ± 0.030	0.821 ± 0.029
IBS (IPCW) $\downarrow$	0.079 ± 0.006	0.073 ± 0.006	0.110 ± 0.013	0.087 ± 0.006	0.067 ± 0.003
MAE $\downarrow$	0.183 ± 0.010	0.089 ± 0.011	0.048 ± 0.005	0.255 ± 0.019	0.067 ± 0.006

Table 4: ALS PRO-ACT: 5-fold cross-validation mean $\pm$ std.

strengthened when the decoder class is better aligned with the frailty-based mechanism than with the independent proxy. In our model, the conditional event and censoring distributions are parameterized as Weibull distributions given (X, Z) . Under dependence, the independent proxy generally induces distorted cause-specific hazards that need not have a simple Weibull form. Consequently, proxy-based explanations may be poorly matched to the admissible decoder class, whereas a frailty-consistent explanation can preserve simple conditional mechanisms and better recover the event-time distribution. In summary, DVFM helps under dependent censoring because its objective can reward latent frailty when it provides enough reconstruction benefit to justify its complexity. Under independent censoring, the same objective does not structurally require shared latent dependence. A fuller objective-based discussion is provided in Appendix B.

7 Experiments

We evaluate DVFM on synthetic benchmarks designed to induce dependent censoring with varying dependence geometry and strength. We compare against DeepSurv (neural Cox model), MTLR (discrete-time survival modeling), and ClaytonAFT (parametric copula-based dependent-censoring model). Each scenario is repeated five times, and we report the mean $\pm$ standard deviation. ClaytonAFT uses Weibull AFT margins for the event and censoring times and a Clayton copula to model their dependence under informative censoring. This baseline follows the general copula-based dependent-censoring framework [12, 13] and is related to recent copula-based deep survival models [3]. We consider three complementary families of metrics. Standard metrics include the concordance index (C-index; higher is better) [14], IBS-IPCW (lower is better) based on the integrated Brier score [15], and MAE (lower is better); a dependence-aware calibration metric, IBS-DEP [16], which evaluates calibration under dependent censoring; and bias decomposition, which involves MAE computed separately on uncensored and censored subsets for synthetic datasets to isolate where performance gains arise. All MAE values were normalized by dividing them by the maximum observed time in the dataset.

7.1 Synthetic benchmark results

Table 1 summarizes the aggregate synthetic benchmark results. We consider 10 different scenarios that have been indicated in Table 2 for generating synthetic data. The values in Table 1 are the average values for all those scenarios with their standard deviations. DVFM achieves the best overall discrimination (C-index) and calibration (IBS-IPCW), and it also attains the best IBS-DEP and MAE Uncensored among the compared methods. Per-scenario results for DVFM and ClaytonAFT are reported in Appendix C. DVFM achieves the strongest overall combination of discrimination and calibration on the synthetic benchmark. Its gains in IBS-IPCW and IBS-DEP indicate improved calibration both under standard evaluation and under informative censoring. ClaytonAFT performs strongly when its assumed copula aligns with the data-generating dependence, particularly on MAE results, but its parametric structure constrains flexibility. DeepSurv and MTLR inherit the independent-censoring assumption and degrade more substantially in the informative-censoring setting.

Metric	CoxPH	DeepSurv	MTLR	ClaytonAFT	DVFM
C-index $\uparrow$	0.611 $\pm$ 0.050	0.609 $\pm$ 0.041	0.584 $\pm$ 0.042	0.635 $\pm$ 0.046	0.638 $\pm$ 0.044
IBS (IPCW) $\downarrow$	0.165 $\pm$ 0.009	0.169 $\pm$ 0.010	0.233 $\pm$ 0.020	0.161 $\pm$ 0.008	0.159 $\pm$ 0.010
MAE $\downarrow$	0.239 $\pm$ 0.017	0.214 $\pm$ 0.016	0.144 $\pm$ 0.012	0.240 $\pm$ 0.015	0.278 $\pm$ 0.022

Table 5: Cancer METABRIC: 5-fold cross-validation mean $\pm$ std.

Metric	CoxPH	DeepSurv	MTLR	ClaytonAFT	DVFM
C-index $\uparrow$	0.749 $\pm$ 0.061	0.804 $\pm$ 0.039	0.756 $\pm$ 0.043	0.791 $\pm$ 0.052	0.822 $\pm$ 0.047
IBS (IPCW) $\downarrow$	0.152 $\pm$ 0.041	0.125 $\pm$ 0.028	0.138 $\pm$ 0.039	0.126 $\pm$ 0.024	0.110 $\pm$ 0.036
MAE $\downarrow$	0.221 $\pm$ 0.091	0.128 $\pm$ 0.021	0.134 $\pm$ 0.029	0.190 $\pm$ 0.039	0.161 $\pm$ 0.040

Table 6: ALS CALSNIC: 5-fold cross-validation mean $\pm$ std.

7.2 Real-world datasets

We evaluate DVFM on three real-world benchmarks with varying censoring regimes: ALS PRO-ACT [17], Cancer METABRIC [18], and ALS CALSNIC [19]. Dataset statistics are reported in Table 3. For ALS PRO-ACT (high censoring), DVFM attains the best C-index and the lowest IBS-IPCW, indicating the strongest overall discrimination and calibration in this heavily censored setting. MTLR achieves the lowest MAE, while DVFM remains highly competitive on pointwise error and strongest on the survival-oriented metrics. For Cancer METABRIC (moderate censoring), DVFM achieves the best IBS-IPCW and the best C-index, while MTLR obtains the lowest MAE. This indicates a more mixed performance landscape than in the synthetic experiments, but DVFM remains strongest in calibration-oriented performance and overall ranking quality. For ALS CALSNIC (small sample, moderate censoring), Despite the small sample size, DVFM achieves the best C-index and the best IBS-IPCW, suggesting that the latent frailty mechanism remains useful even in limited-data settings. DeepSurv attains the lowest MAE, indicating that pointwise error and survival calibration need not always rank methods in the same order.

8 Limitations

First, the method relies on the shared latent-frailty assumption $E \perp C \mid (X, Z)$, which may be misspecified if dependence arises from mechanisms not captured by a common latent variable. Second, the current implementation uses Weibull conditional distributions for both event and censoring times, which may be restrictive for more complex hazard shapes. Third, the theoretical results are structural and model-class dependent; they justify when the C-ELBO can prefer frailty-based explanations, but they do not guarantee universal recovery of the true event-time distribution under arbitrary dependent censoring.

9 Conclusion

We introduced the Deep Variational Frailty Model (DVFM), a deep generative approach for survival analysis under dependent censoring. By modeling event and censoring times through a shared latent frailty variable, DVFM provides a flexible alternative to methods that either assume independent censoring or require a prespecified copula family. Our formulation combines joint modeling of event and censoring mechanisms with amortized variational inference and a censored ELBO objective, yielding a practical framework for end-to-end training under right censoring. From a predictive perspective, this objective provides a useful inductive bias: it does not encourage unnecessary latent structure when censoring is independent, but under dependent censoring it can favor frailty-based representations that better preserve the event-time distribution. Across synthetic and real-world datasets, DVFM achieved strong discrimination and calibration, with the clearest gains appearing in settings where dependent censoring is more likely to matter. These results suggest that latent shared heterogeneity is a promising modeling principle for extending survival prediction beyond the standard independent-censoring setting.

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A Derivation of the Censored ELBO

We derive the ELBO for right-censored observations (t, δ, x) . Under the generative model

$$p_\theta(e, c, z | x) = p_\theta(e | x, z)p_\theta(c | x, z)p(z), \quad (21)$$

we observe (t, δ) where $t = \min(e, c)$ and $\delta = \mathbb{I}(e \leq c)$.

Observed likelihood given z . If $\delta = 1$, the event occurs at time t and censoring exceeds t :

$$p(t, \delta = 1 | x, z) = f_E(t | x, z)S_C(t | x, z). \quad (22)$$

If $\delta = 0$, censoring occurs at time t and the event exceeds t :

$$p(t, \delta = 0 | x, z) = S_E(t | x, z)f_C(t | x, z). \quad (23)$$

Combining both cases yields (11).

ELBO. For any variational posterior $q_\phi(z | t, \delta, x)$,

$$\log p_\theta(t, \delta | x) = \log \int p_\theta(t, \delta, z | x) dz \quad (24)$$

$$= \log \int q_\phi(z | t, \delta, x) \frac{p_\theta(t, \delta, z | x)}{q_\phi(z | t, \delta, x)} dz \quad (25)$$

$$\geq \mathbb{E}_{q_\phi}[\log p_\theta(t, \delta, z | x) - \log q_\phi(z | t, \delta, x)] \quad (26)$$

$$= \mathbb{E}_{q_\phi}[\log p_\theta(t, \delta | x, z)] - D_{\text{KL}}(q_\phi(z | t, \delta, x) \| p(z)), \quad (27)$$

which yields (12). In practice, we optionally use β -annealing and free bits to stabilize optimization.

B Objective-Based Perspective on the Predictive Behavior of DVFM

This appendix provides an objective-based explanation for the empirical behavior of DVFM under independent and dependent censoring. The purpose is not to reinterpret DVFM as a formal test for dependent censoring, nor to claim unrestricted identification of the joint distribution of event and censoring times. Such unrestricted identification is impossible from right-censored observations without additional structural assumptions, as discussed in the main text. Instead, the goal is to clarify how the censored evidence lower bound (C-ELBO) interacts with the structural bias imposed by the shared latent frailty model.

The key idea is the following. Under independent censoring, the observed data can be explained without a shared latent dependence mechanism. In this case, the reconstruction term of the C-ELBO does not require the posterior to encode additional information through the frailty variable, while the KL term penalizes unnecessary deviations of the variational posterior from the prior. Thus, the objective does not structurally reward spurious dependence. Under dependent censoring, however, an independence-based explanation may correspond to a Tsatis-type proxy whose event and censoring hazards are distorted relative to the true marginal mechanisms. If this proxy is poorly matched to the decoder class, while the true dependent mechanism is well represented by a frailty-based model, then the C-ELBO can favor a nontrivial latent frailty representation whenever the improvement in reconstruction outweighs the additional KL cost.

B.1 Preliminaries

Let E denote the event time, C the censoring time, X the observed covariates, and

$$T = \min(E, C), \quad \delta = \mathbb{I}\{E \leq C\}.$$

For simplicity, the arguments below are written at the population level and conditional on a fixed covariate value $X = x$; the same reasoning applies pointwise in x or after averaging over the covariate distribution.

The observed-data likelihood contribution is

$$L_{\text{joint}}(t, \delta | x, z) = [f_E(t | x, z)S_C(t | x, z)]^\delta [S_E(t | x, z)f_C(t | x, z)]^{1-\delta}.$$

369 The negative C-ELBO objective can be written as

$$\mathcal{J}(\theta, \phi) = -\mathbb{E}_{q_\phi(z|t, \delta, x)} [\log L_{\text{joint}}(t, \delta | x, z)] + \beta D_{\text{KL}}(q_\phi(z | t, \delta, x) \| p(z)).$$

370 The first term measures observed-data reconstruction, while the second term regularizes the amount
371 of information encoded in the latent frailty posterior.

372 We will use the following terminology.

373 **Tsiatis independent proxy.** For any observed competing-risk distribution of (T, δ) , one can define
374 an independent proxy distribution P^* whose event and censoring hazards match the observed cause-
375 specific hazards. Let

$$h_1(t | x) = \lim_{\Delta t \downarrow 0} \frac{\Pr(t \leq T < t + \Delta t, \delta = 1 | T \geq t, X = x)}{\Delta t}$$

376 and

$$h_0(t | x) = \lim_{\Delta t \downarrow 0} \frac{\Pr(t \leq T < t + \Delta t, \delta = 0 | T \geq t, X = x)}{\Delta t}.$$

377 The Tsiatis independent proxy is the independent model with

$$h_E^*(t | x) = h_1(t | x), \quad h_C^*(t | x) = h_0(t | x).$$

378 This proxy induces the same distribution of the observed data (T, δ) , even when the true joint
379 distribution of (E, C) is dependent.

380 **Frailty-based model class.** DVFM restricts the joint distribution through the conditional indepen-
381 dence structure

$$E \perp C | (X, Z),$$

382 with conditional event and censoring distributions parameterized by the decoder. In the experiments
383 of this paper, these conditional distributions are Weibull:

$$E | X, Z \sim \text{Weibull}(k_E(X, Z), \lambda_E(X, Z)), \quad C | X, Z \sim \text{Weibull}(k_C(X, Z), \lambda_C(X, Z)).$$

384 After marginalizing over Z , the resulting marginal event and censoring distributions need not be
385 Weibull; they are mixtures induced by the latent frailty distribution. Thus, the relevant model
386 class is not the single Weibull family, but the class of observed-data distributions induced by latent
387 marginalization of conditional Weibull decoders.

388 **Population reconstruction loss.** For a candidate parameter pair (θ, ϕ) , define the population
389 reconstruction loss

$$\mathcal{R}(\theta, \phi) = -\mathbb{E}_{P_0(t, \delta, x)} \mathbb{E}_{q_\phi(z|t, \delta, x)} [\log L_{\text{joint}}(t, \delta | x, z)],$$

390 where P_0 denotes the true observed-data distribution. Then

$$\mathcal{J}(\theta, \phi) = \mathcal{R}(\theta, \phi) + \beta \mathbb{E}_{P_0(t, \delta, x)} D_{\text{KL}}(q_\phi(z | t, \delta, x) \| p(z)).$$

391 This decomposition is the basis of the objective-based comparison below.

392 B.2 Behavior under Independent Censoring

393 We first consider the case where the standard independent-censoring assumption is correct:

$$E \perp C | X.$$

394 **Proposition 2** (Objective behavior under independent censoring). *Assume that $E \perp C | X$ and that*
395 *the true independent event and censoring mechanisms are representable within the DVFM decoder*
396 *class by a parameter setting in which the shared latent frailty is not needed for reconstruction. Then an*
397 *independence-based explanation can attain the optimal population reconstruction term. Consequently,*
398 *any solution that uses an informative posterior over Z is preferred by the C-ELBO only if its*
399 *reconstruction improvement compensates for its additional KL cost. In particular, under independent*
400 *censoring, the C-ELBO does not structurally require a nontrivial shared frailty representation.*

401 *Proof.* Under conditional independence,

$$\Pr(E > t, C > t \mid X = x) = S_E(t \mid x)S_C(t \mid x).$$

402 The observed cause-specific event hazard is therefore

$$h_1(t \mid x) = \frac{f_E(t \mid x)S_C(t \mid x)}{S_E(t \mid x)S_C(t \mid x)} = h_E(t \mid x),$$

403 and similarly,

$$h_0(t \mid x) = h_C(t \mid x).$$

404 Thus, when censoring is truly independent, the Tsiatis independent proxy coincides with the true
405 independent event and censoring mechanisms:

$$h_E^*(t \mid x) = h_E(t \mid x), \quad h_C^*(t \mid x) = h_C(t \mid x).$$

406 By assumption, this independent explanation is representable within the DVFM decoder class without
407 requiring the latent variable to carry additional subject-specific information about the dependence
408 between E and C . Therefore, the population reconstruction loss can already be minimized by
409 a solution in which the shared frailty is redundant for explaining the observed event–censoring
410 relationship.

411 Now consider another solution that uses an informative variational posterior $q_\phi(z \mid t, \delta, x)$ to encode
412 additional latent structure. Such a solution may be selected only if the reduction in reconstruction
413 loss is large enough to offset the KL term

$$\beta D_{\text{KL}}(q_\phi(z \mid t, \delta, x) \parallel p(z)).$$

414 When the independent explanation already achieves the optimal reconstruction term, there is no
415 structural reconstruction advantage that requires such additional latent information. Hence the C-
416 ELBO does not intrinsically reward spurious frailty-based dependence under independent censoring. $\square$
417

418 **Remark 2.** *Proposition 2 does not claim that the learned latent variable is exactly inactive in every*
419 *finite-sample or nonconvex optimization run. Neural decoders may still use Z for approximation,*
420 *optimization, or finite-sample reasons. The claim is only that, at the population objective level,*
421 *independent censoring does not require shared latent dependence to explain the observed-data*
422 *likelihood.*

423 B.3 Behavior under Dependent Censoring

424 We now consider the case where censoring is informative:

$$E \not\perp C \mid X.$$

425 In this setting, the observed cause-specific hazards generally differ from the true marginal event
426 and censoring hazards. Consequently, the Tsiatis independent proxy can reproduce the observed
427 distribution of (T, δ) while distorting the marginal event and censoring mechanisms. This is the
428 source of bias for prediction of the event-time distribution.

429 To make the objective comparison precise, let $\mathcal{F}_{\text{frailty}}$ denote the subset of DVFM parameterizations
430 that use the shared latent variable nontrivially, and let $\mathcal{F}_{\text{proxy}}$ denote the subset of admissible
431 parameterizations that attempt to represent the observed data through an independence-based proxy
432 explanation. Define the best population reconstruction losses

$$\mathcal{R}_{\text{frailty}}^* = \inf_{(\theta, \phi) \in \mathcal{F}_{\text{frailty}}} \mathcal{R}(\theta, \phi),$$

433 and

$$\mathcal{R}_{\text{proxy}}^* = \inf_{(\theta, \phi) \in \mathcal{F}_{\text{proxy}}} \mathcal{R}(\theta, \phi).$$

434 We say that the proxy class has a positive reconstruction gap if

$$\epsilon = \mathcal{R}_{\text{proxy}}^* - \mathcal{R}_{\text{frailty}}^* > 0.$$

435 This assumption formalizes the case where the frailty-based explanation is better matched to the
436 decoder class than the independent proxy.

Proposition 3 (Objective preference under dependent censoring). *Assume that the observed-data distribution is generated by a dependent censoring mechanism that is representable by a frailty-based DVFM model. Suppose further that every admissible independence-proxy explanation has a positive population reconstruction gap*

$$\epsilon = \mathcal{R}_{\text{proxy}}^* - \mathcal{R}_{\text{frailty}}^* > 0.$$

Let

$$\Delta_{\text{KL}} = \mathbb{E}_{P_0} D_{\text{KL}}(q_{\phi_f}(z | t, \delta, x) \| p(z)) - \mathbb{E}_{P_0} D_{\text{KL}}(q_{\phi_p}(z | t, \delta, x) \| p(z))$$

denote the additional KL cost of a frailty-based solution relative to a proxy-based solution. If

$$\epsilon > \beta \Delta_{\text{KL}},$$

then the population C-ELBO objective is smaller for the frailty-based solution than for the proxy-based solution. Hence, in this regime, optimization of the C-ELBO favors a nontrivial frailty representation.

Proof. Let (θ_f, ϕ_f) be a frailty-based solution and (θ_p, ϕ_p) be a proxy-based solution. Their population negative C-ELBO values are

$$\mathcal{J}_f = \mathcal{R}(\theta_f, \phi_f) + \beta \mathbb{E}_{P_0} D_{\text{KL}}(q_{\phi_f}(z | t, \delta, x) \| p(z)),$$

and

$$\mathcal{J}_p = \mathcal{R}(\theta_p, \phi_p) + \beta \mathbb{E}_{P_0} D_{\text{KL}}(q_{\phi_p}(z | t, \delta, x) \| p(z)).$$

By the positive reconstruction-gap assumption,

$$\mathcal{R}(\theta_p, \phi_p) - \mathcal{R}(\theta_f, \phi_f) \geq \epsilon.$$

Therefore,

$$\mathcal{J}_p - \mathcal{J}_f = [\mathcal{R}(\theta_p, \phi_p) - \mathcal{R}(\theta_f, \phi_f)] - \beta \Delta_{\text{KL}} \geq \epsilon - \beta \Delta_{\text{KL}}.$$

If $\epsilon > \beta \Delta_{\text{KL}}$, then

$$\mathcal{J}_p - \mathcal{J}_f > 0,$$

so the frailty-based solution has a smaller negative C-ELBO objective. Hence the C-ELBO favors the frailty-based explanation in this regime. $\square$

Remark 3. *The positive reconstruction-gap condition is a model-class condition, not a universal property of dependent censoring. It holds when the independent proxy is poorly approximated by the admissible decoder class, while the dependent mechanism is well approximated by a frailty-based parameterization. This condition should be verified empirically or treated as an inductive-bias assumption.*

B.4 Why the Independent Proxy Can Be Mismatched to the Decoder Class

We now give intuition for why the positive reconstruction-gap condition in Proposition 3 can arise.

Under dependent censoring, the observed cause-specific event hazard is generally not equal to the true marginal event hazard. Instead, it can often be written in the form

$$h_1(t | x) = h_E(t | x) A_E(t | x),$$

where $A_E(t | x)$ is a distortion factor induced by the dependence between E and C . Similarly,

$$h_0(t | x) = h_C(t | x) A_C(t | x).$$

When dependence is generated by a copula or by shared frailty, these distortion factors are generally nonconstant functions of time and covariates. Therefore, even if the true conditional event and censoring hazards have a simple parametric form given the latent frailty, the observed cause-specific hazards of the independent proxy may have a more complicated shape.

For example, suppose the dependence between E and C is represented through a survival copula $\bar{C}(u, v)$, where

$$u = S_E(t | x), \quad v = S_C(t | x).$$

Then the joint survival function is

$$\Pr(E > t, C > t | x) = \bar{C}(u, v).$$

471 The observed cause-specific event hazard can be written as

$$h_1(t | x) = h_E(t | x) \frac{u \partial_u \bar{C}(u, v)}{\bar{C}(u, v)} \Big|_{u=S_E(t|x), v=S_C(t|x)}.$$

472 Thus the distortion factor is

$$A_E(t | x) = \frac{S_E(t | x) \partial_u \bar{C}(u, v)}{\bar{C}(u, v)} \Big|_{u=S_E(t|x), v=S_C(t|x)}.$$

473 Except in special cases, $A_E(t | x)$ is not constant and does not preserve the original hazard family.
474 Hence the independent proxy hazard

$$h_E^*(t | x) = h_1(t | x)$$

475 may lie outside, or be poorly approximated by, the decoder class used by the model.

476 This observation explains why DVFM can be useful even though the observed-data likelihood is
477 nonidentifiable in an unrestricted sense. The model does not identify the true joint distribution without
478 assumptions. Rather, it imposes a structural preference: dependence should be explained through a
479 shared latent frailty that preserves simple conditional event and censoring mechanisms, rather than
480 through arbitrary distorted marginal hazards.

481 B.5 Implication for Prediction

482 The preceding propositions explain the predictive behavior of DVFM as an objective-level phe-
483 nomenon.

484 Under independent censoring, the event and censoring processes can be modeled separately without
485 loss of information for the event-time distribution. If this independent explanation is contained in the
486 decoder class, DVFM has no population-level need to encode a shared frailty dependence. The KL
487 term therefore acts as a regularizer against unnecessary posterior information.

488 Under dependent censoring, the independent proxy can reproduce the observed distribution of (T, δ) ,
489 but it may do so using distorted event and censoring hazards. These distorted hazards can be
490 unfavorable for recovering the event-time marginal distribution, especially when the decoder class
491 is better aligned with the true frailty-based mechanism than with the proxy. In that case, the joint
492 reconstruction term can improve by using a nontrivial latent frailty variable. When this gain is larger
493 than the additional KL cost, the C-ELBO favors the frailty-based explanation.

494 This argument should be interpreted as an inductive-bias explanation rather than a general identifiability
495 theorem. DVFM can improve prediction when its structural assumptions are aligned with the
496 data-generating mechanism, but the method does not eliminate the fundamental nonidentifiability of
497 dependent censoring from right-censored observations alone.

498 C Additional Synthetic Scenario Tables

499 For transparency, Tables 7 and 8 report per-scenario synthetic results provided in the draft materials.

500 C.1 Sensitivity to the Latent Frailty Dimension

501 We further examine the sensitivity of DVFM to the dimension d of the latent frailty variable $z \in \mathbb{R}^d$.
502 Since d controls the capacity of the shared latent representation, too small a value may under-
503 represent unobserved heterogeneity, whereas too large a value may introduce unnecessary complexity.
504 We evaluate $d \in \{2, 5, 10, 20\}$ on three real datasets and on synthetic datasets generated from five
505 dependence families with different dependence strengths. The results are reported as mean $\pm$ standard
506 deviation over the corresponding cross-validation folds or synthetic settings.

507 Table 9 shows the real-data results. Overall, DVFM is not highly sensitive to the choice of d . On ALS
508 PRO-ACT, increasing d from 2 to 20 slightly improves IBS-IPCW and MAE, while the best C-index
509 is obtained at $d = 5$. On Cancer METABRIC, the results are similarly stable across latent dimensions,
510 with $d = 20$ giving the lowest IBS-IPCW and the best C-index, and $d = 2$ giving the lowest MAE. On
511 ALS CALSNIC, performance improves more clearly as d increases: the best C-index and IBS-IPCW

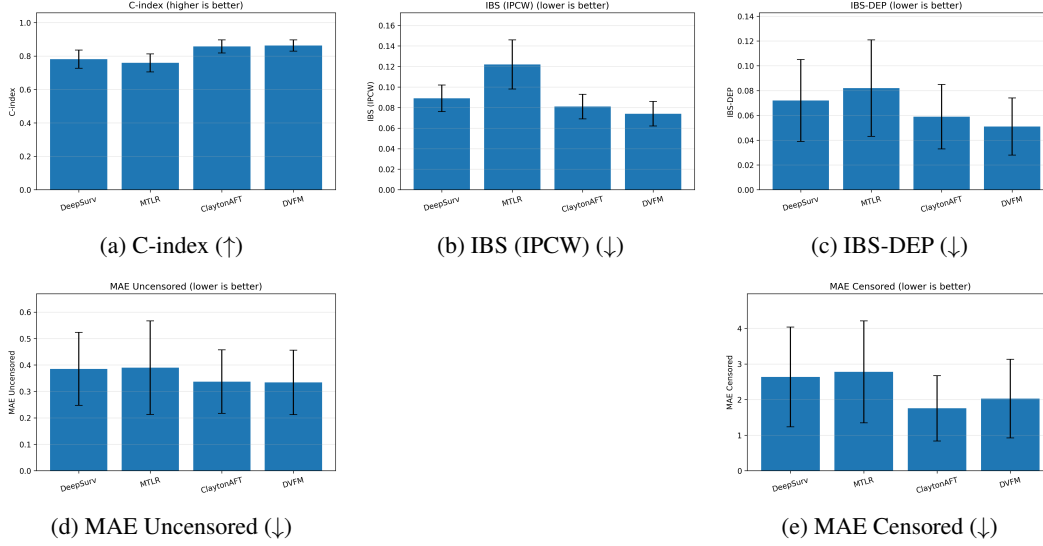


Figure 3: Comparison of DeepSurv, MTLR, ClaytonAFT, and DVFM on the synthetic benchmark. Bars show the mean over 5 runs and error bars indicate one standard deviation.

Scenario	C-index	IBS-IPCW	MAE	IBS-DEP
S1	0.877 ± 0.038	0.069 ± 0.006	0.302 ± 0.123	0.035 ± 0.022
S2	0.892 ± 0.040	0.064 ± 0.004	0.298 ± 0.129	0.031 ± 0.020
S3	0.743 ± 0.043	0.124 ± 0.021	0.372 ± 0.069	0.077 ± 0.022
S4	0.760 ± 0.045	0.127 ± 0.023	0.624 ± 0.175	0.028 ± 0.012
S5	0.862 ± 0.041	0.076 ± 0.011	0.364 ± 0.120	0.024 ± 0.014
S6	0.878 ± 0.036	0.009 ± 0.011	0.265 ± 0.082	0.121 ± 0.075
S7	0.861 ± 0.039	0.089 ± 0.015	0.318 ± 0.136	0.038 ± 0.024
S8	0.877 ± 0.037	0.084 ± 0.016	0.330 ± 0.143	0.032 ± 0.017
S9	0.962 ± 0.007	0.031 ± 0.003	0.232 ± 0.103	0.082 ± 0.019
S10	0.919 ± 0.016	0.065 ± 0.009	0.267 ± 0.117	0.045 ± 0.004

Table 7: DVFM performance across 10 synthetic scenarios.

are obtained at $d = 20$, suggesting that a higher-dimensional latent frailty can better capture the underlying dependence structure in this dataset. However, the lowest MAE occurs at $d = 2$, again indicating that ranking/calibration metrics and pointwise error may favor different model capacities.

Table 10 summarizes the sensitivity analysis on the synthetic benchmarks, averaged over the five dependence families and the considered dependence strengths. The synthetic results show very small variation across latent dimensions. The C-index remains around 0.83 for all values of d , and IBS-IPCW changes only marginally. The best average C-index is obtained at $d = 20$, while the best IBS-IPCW is obtained at $d = 5$. For subset-wise MAE, $d = 10$ gives the lowest uncensored and censored errors. These results suggest that DVFM does not require careful tuning of the latent dimension to remain effective; moderate latent dimensions such as $d = 5$ or $d = 10$ are generally sufficient, while $d = 20$ can be useful when the dependence structure is more complex.

Overall, the sensitivity analysis indicates that DVFM is robust to the latent dimension. The model does not exhibit a strong degradation when d is small, and larger values of d provide only modest improvements except in settings where additional latent capacity appears beneficial. In the experiments considered here, $d = 10$ provides a good default trade-off between predictive accuracy, calibration, and model complexity.

Scenario	C-index	IBS-IPCW	MAE	IBS-DEP
S1	0.873 ± 0.043	0.078 ± 0.006	0.290 ± 0.120	0.041 ± 0.027
S2	0.882 ± 0.045	0.075 ± 0.004	0.299 ± 0.127	0.041 ± 0.030
S3	0.745 ± 0.046	0.125 ± 0.021	0.360 ± 0.055	0.079 ± 0.023
S4	0.763 ± 0.046	0.127 ± 0.021	0.639 ± 0.177	0.031 ± 0.015
S5	0.859 ± 0.047	0.084 ± 0.010	0.373 ± 0.119	0.026 ± 0.015
S6	0.839 ± 0.064	0.009 ± 0.010	0.278 ± 0.055	0.143 ± 0.077
S7	0.862 ± 0.039	0.093 ± 0.014	0.309 ± 0.134	0.039 ± 0.024
S8	0.875 ± 0.038	0.089 ± 0.014	0.329 ± 0.146	0.034 ± 0.022
S9	0.959 ± 0.009	0.060 ± 0.012	0.302 ± 0.174	0.085 ± 0.018
S10	0.914 ± 0.015	0.073 ± 0.009	0.262 ± 0.115	0.067 ± 0.010

Table 8: ClaytonAFT performance across 10 synthetic scenarios.

Table 9: Sensitivity of DVFM to the latent frailty dimension d on real datasets. Results are reported as mean $\pm$ standard deviation. Best values for each dataset and metric are shown in bold.

Dataset	d	C-index $\uparrow$	IBS-IPCW $\downarrow$	MAE $\downarrow$
ALS PRO-ACT	2	0.805 ± 0.045	0.063 ± 0.009	0.063 ± 0.005
	5	0.815 ± 0.043	0.063 ± 0.010	0.065 ± 0.007
	10	0.803 ± 0.038	0.062 ± 0.011	0.056 ± 0.006
	20	0.808 ± 0.030	0.061 ± 0.013	0.061 ± 0.006
Cancer METABRIC	2	0.655 ± 0.012	0.161 ± 0.008	0.266 ± 0.023
	5	0.645 ± 0.017	0.164 ± 0.006	0.286 ± 0.019
	10	0.656 ± 0.019	0.160 ± 0.007	0.275 ± 0.027
	20	0.656 ± 0.015	0.159 ± 0.006	0.279 ± 0.018
ALS CALSNIC	2	0.586 ± 0.082	0.162 ± 0.034	0.140 ± 0.035
	5	0.645 ± 0.121	0.156 ± 0.038	0.163 ± 0.057
	10	0.773 ± 0.074	0.120 ± 0.032	0.166 ± 0.046
	20	0.833 ± 0.057	0.115 ± 0.035	0.171 ± 0.050

C.2 Sensitivity to the Maximum KL Weight

We also study the sensitivity of DVFM to the maximum value of the KL-regularization weight, denoted by $\beta_{\max}$, in the C-ELBO objective. In our implementation, β is gradually annealed during training and capped at $\beta_{\max}$. This parameter controls the trade-off between joint reconstruction of the event and censoring mechanisms and regularization of the latent frailty posterior. A small $\beta_{\max}$ allows the model to use the latent frailty variable more freely, whereas a larger $\beta_{\max}$ imposes stronger pressure toward the prior and may reduce unnecessary latent complexity.

Table 11 reports the sensitivity results on three real datasets for $\beta_{\max} \in \{0.2, 0.5, 1.0\}$. On ALS PRO-ACT, increasing $\beta_{\max}$ improves all reported metrics, with $\beta_{\max} = 1.0$ achieving the best C-index, IBS-IPCW, and MAE. This suggests that, in the highly censored PRO-ACT setting, stronger latent regularization improves both discrimination and calibration while reducing prediction error. On Cancer METABRIC, performance is relatively stable across the three values. The best C-index is obtained at $\beta_{\max} = 0.5$, while $\beta_{\max} = 0.2$ gives the best IBS-IPCW, and $\beta_{\max} = 1.0$ gives the lowest MAE. On ALS CALSNIC, $\beta_{\max} = 1.0$ yields the best C-index and IBS-IPCW, while $\beta_{\max} = 0.2$ gives the lowest MAE. Overall, the results indicate that DVFM is reasonably robust to $\beta_{\max}$, with $\beta_{\max} = 1.0$ providing a strong default choice for ranking and calibration metrics, while smaller values may be preferable when pointwise error is prioritized.

Table 10: Sensitivity of DVFM to the latent frailty dimension d on synthetic datasets averaged over five dependence families and different dependence strengths. Results are reported as mean $\pm$ standard deviation.

d	C-index $\uparrow$	IBS-IPCW $\downarrow$	MAE Uncensored $\downarrow$	MAE Censored $\downarrow$
2	0.828 $\pm$ 0.092	0.094 $\pm$ 0.054	0.447 $\pm$ 0.155	1.910 $\pm$ 1.023
5	0.830 $\pm$ 0.088	0.093 $\pm$ 0.052	0.450 $\pm$ 0.151	1.957 $\pm$ 1.049
10	0.830 $\pm$ 0.089	0.094 $\pm$ 0.053	0.445 $\pm$ 0.150	1.897 $\pm$ 1.010
20	0.831 $\pm$ 0.091	0.093 $\pm$ 0.052	0.447 $\pm$ 0.157	1.919 $\pm$ 1.056

Table 11: Sensitivity of DVFM to the maximum KL weight $\beta_{\max}$ on real datasets. Results are reported as mean $\pm$ standard deviation. Best values for each dataset and metric are shown in bold.

Dataset	$\beta_{\max}$	C-index $\uparrow$	IBS-IPCW $\downarrow$	MAE $\downarrow$
ALS PRO-ACT	0.2	0.780 $\pm$ 0.023	0.085 $\pm$ 0.004	0.149 $\pm$ 0.030
	0.5	0.803 $\pm$ 0.020	0.075 $\pm$ 0.004	0.114 $\pm$ 0.026
	1.0	0.826 $\pm$ 0.015	0.066 $\pm$ 0.004	0.076 $\pm$ 0.017
Cancer METABRIC	0.2	0.654 $\pm$ 0.020	0.155 $\pm$ 0.006	0.284 $\pm$ 0.034
	0.5	0.655 $\pm$ 0.018	0.156 $\pm$ 0.006	0.277 $\pm$ 0.024
	1.0	0.655 $\pm$ 0.018	0.158 $\pm$ 0.006	0.259 $\pm$ 0.020
ALS CALSNC	0.2	0.756 $\pm$ 0.052	0.140 $\pm$ 0.033	0.140 $\pm$ 0.024
	0.5	0.769 $\pm$ 0.058	0.136 $\pm$ 0.038	0.144 $\pm$ 0.025
	1.0	0.802 $\pm$ 0.059	0.135 $\pm$ 0.038	0.144 $\pm$ 0.025

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